

# Pei Zhang

✉ [pei.zhang@nih.gov](mailto:pei.zhang@nih.gov)

🌐 <https://peizhang.org>

🎓 [Google Scholar](#)

🐙 [GitHub](#)

📍 9609 Medical Center Dr, Room 7E614, Rockville, MD 20850, USA

☎ +1 (240) 992-7823

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## Education

- 2019/09 - 2025/05     **Ph.D. in Mathematical Statistics, University of Maryland, College Park (UMD)**  
Dissertation title: *Mixed Modeling Approaches for Characterizing Genetic Effects and Heritability Metrics in Longitudinal Phenotypes* (NIH Outstanding Graduate Research)  
Primary Advisors: Drs. Paul S. Albert (NIH Biostatistics) and Doron Levy (UMD Applied Mathematics)  
Co-advisors: Drs. Jianxin Shi (NIH Biostatistics) and Hyokyoung G. Hong (NIH Biostatistics)
- 2014/09 - 2016/06     **M.S. in Mathematics, University of Science and Technology of China (USTC)**  
Dissertation title: *Ekeland Index Theory for Periodic Solutions and Symmetric Periodic Solutions of Convex Hamiltonian Systems*  
Advisor: Dr. Xinan Ma  
Co-advisor: Dr. Hui Liu
- 2010/09 - 2014/06     **B.S. in Mathematics and Applied Mathematics (Teacher Education Program), Nanjing Normal University (NNU)**  
Dissertation title: *Some Problems in Hamiltonian Systems* (NNU Outstanding Graduation Thesis)  
Advisor: Dr. Yujun Dong
- 2009/09 - 2010/06     **Undergraduate in Computer Science, NNU**

## Employment History

- 2025/05 - present     **Postdoctoral Fellow (Mentored by Drs. Paul S. Albert and Jianxin Shi)**  
*Biostatistics Branch (BB), Division of Cancer Epidemiology & Genetics (DCEG)  
National Cancer Institute (NCI), National Institutes of Health (NIH)*
- 2021/09 - 2025/05     **Predoctoral Fellow (Mentored by Drs. Paul S. Albert and Jianxin Shi, and Hyokyoung G. Hong)**  
*BB, DCEG  
NCI, NIH*
- 2019/08 - 2021/08     **Teaching Assistant**  
*Department of Mathematics, UMD*
- 2016/08 - 2017/04     **Mathematics Teacher**  
*NO.1 Middle School Affiliated to Central China Normal University, Wuhan, China*

## Employment History (continued)

2014/09 - 2016/06

■ **Teaching Assistant**  
*Department of Mathematics, USTC*

## Research Interests

- **Longitudinal and correlated data methodology**, with emphasis on repeated-measures analysis, biomarker studies, and complex dependent data structures.
- **Statistical and computational methods in genetics** for analyzing electronic health records (EHR), real-world data (RWD), and large-scale population studies (e.g., the Prostate, Lung, Colorectal, and Ovarian (PLCO) Cancer Screening Trial), with applications to prostate cancer risk stratification, early detection, and prediction.
- **Statistical and machine learning approaches in epidemiology**, including machine learning and deep learning for cancer genetic epidemiology, nutritional epidemiology, and metabolomics.

## Publications

\* corresponding author    † co-first / equal contribution    ‡ alphabetical order

### Methodological / Statistical Papers

- 1 Bandreddi, Sumaja<sup>†</sup>, **Zhang, Pei**<sup>†\*</sup>, R. Kelly, M. Machiela, and Albert, Paul S.<sup>\*</sup>, “Analysis of semi-continuous longitudinal data using tobit and two-part hurdle models with application to clonal hematopoiesis,” To be submitted to *Statistics in Medicine*, 2026.
- 2 **Zhang, Pei**, X. Wang, Shi, Jianxin<sup>\*</sup>, and Albert, Paul S.<sup>\*</sup>, “Estimating the heritability of longitudinal rate-of-change: Genetic insights into PSA velocity in prostate cancer-free individuals,” 2026, *Biostatistics* (In Press), preprint on arXiv [[arXiv](#)].
- 3 **Zhang, Pei**, Albert, Paul S.<sup>\*</sup>, and Hong, Hyokyung G.<sup>\*</sup>, “Mixed modeling approach for characterizing the genetic effects in a longitudinal phenotype,” *Annals of Applied Statistics*, vol. 19, no. 3, pp. 2070–2087, 2025, [[PDF](#)].

### Collaborative / Applied Papers

- 1 H. Fan, H. Zhao, **Zhang, Pei**, P. Yu, Y. Ji, G. Chen, H. Jin, Y. Liu, J. Liu, Z.-S. Chen, Lyu, Aiping<sup>\*</sup>, Liang, Xinmiao<sup>\*</sup>, and Chen, Yang<sup>\*</sup>, “Homoisoflavanone delays colorectal cancer progression via DNA damage-induced mitochondrial apoptosis and parthanatos-like cell death,” *Advanced Science*, e11406, 2026, [[PDF](#)].
- 2 H. Fan, H. Zhao, L. Gao, Y. Dong, **Zhang, Pei**, P. Yu, Y. Ji, Z.-S. Chen, X. Liang, and Chen, Yang<sup>\*</sup>, “CCN1 enhances tumor immunosuppression through collagen-mediated chemokine secretion in pancreatic cancer,” *Advanced Science*, e2500589, 2025, [[PDF](#)].
- 3 Loftfield, Erika<sup>†\*</sup>, **Zhang, Pei**<sup>†</sup>, C. P. O’Connell, L. L. Kahle, K. Herrick, L. Abar, N. Khandpur, E. M. Steele, and H. G. Hong, “Performance of a food frequency questionnaire for estimating ultraprocessed food intake according to the NOVA classification system in the United States NIH-AARP diet and health study,” *Journal of Nutrition*, vol. 155, no. 7, pp. 2376–2384, 2025, [[PDF](#)].
- 4 Wu, Yetian<sup>†</sup>, **Zhang, Pei**<sup>†</sup>, H. Fan, C. Zhang, P. Yu, X. Liang, and Chen, Yang<sup>\*</sup>, “GPR35 acts a dual role and therapeutic target in inflammation,” *Frontiers in Immunology*, vol. 14, no. 1254446, 2023, [[PDF](#)].

## Preprints / Under Review

- 1 Lim, Jungeun, **Zhang, Pei**, J. Rhee, M. C. Playdon, A. J. Cross, R. Stolzenberg-Solomon, V. L. Roger, M. P. Purdue, Albanes, Demetrius\*, Hong, Hyokyoung G.\*, and Wong, Jason Y. Y.\*, “Metabolomic profile score for predicting serum per- and polyfluoroalkyl substance (PFAS) levels: A novel application of machine learning to enhance detection of environmental pollutants,” Under review at *International Journal of Hygiene and Environmental Health*, 2026.
- 2 M. Zhang, X. Zhang, L. Wang, X. Liu, X. Zhang, J. Li, K. Wang, H. Li, H. Koka, T. Ma, W. Luo, S. Chavez, Y. Kim, A. Morales Miranda, T. Veith, **Zhang, Pei**, W. Zhou, T. Zhang, X. Yang, Y. Zhang, T. Zhang, and J. Bai, “A single-cell atlas of chordoma reveals a primitive malignant state linked to immune remodeling and clinical risk,” Under review at *Cancer Discovery*, 2026.

## PhD Thesis

- 1 **Zhang, Pei**, “Mixed modeling approaches for characterizing genetic effects and heritability metrics in longitudinal phenotypes,” [\[PDF\]](#), Ph.D. dissertation, University of Maryland, College Park, 2025.

## Honors

|             |                                                                                                                                                                                                                                                                                                                                                                                                              |
|-------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2026        |  Finalist (one of six selected finalists) for the IBS/CWS Florence Nightingale Award, International Biometric Conference 2026, Seoul, South Korea                                                                                                                                                                           |
|             |  NCI Early K99/R00 Grant Application — Rated as “Competitive, Not Discussed”                                                                                                                                                                                                                                                |
|             |  Certificate of Completion of Easylearning NLP GenAI Agent Course                                                                                                                                                                                                                                                          |
|             |  National Science Foundation (NSF) Travel Support Award, USA                                                                                                                                                                                                                                                              |
|             |  Institute of Mathematical Statistics (IMS) New Researcher Travel Award, USA                                                                                                                                                                                                                                              |
| 2025 & 2026 |  NIH Postdoctoral Fellowship (twice)                                                                                                                                                                                                                                                                                      |
| 2025        |  NCI-DCEG Institutional Nominee for NCI Pathway to Independence Award for Outstanding Early Stage Postdoctoral Researchers (K99/R00) <a href="#">[about "Early K99" Grant - NIH]</a>   <a href="#">[about "Early K99" Grant - Stanford Medicine]</a>   <a href="#">[Notice of Funding Opportunity Number: PAR-23-286]</a> |
|             |  NIH Fellows Award for Research Excellence (FARE) 2026 (top 25% of all applicants) <a href="#">[List of FARE 2026 Awardees Abstracts]</a>   <a href="#">[about FARE]</a>                                                                                                                                                  |
|             |  NIH Outstanding Graduate Research                                                                                                                                                                                                                                                                                        |
| 2023 & 2024 |  Women in Statistics and Data Science Conference Student Travel Award (twice)                                                                                                                                                                                                                                             |
| 2023        |  UMD Jacob K. Goldhaber Travel Grant                                                                                                                                                                                                                                                                                      |
|             |  UMD International Conference Student Support Award                                                                                                                                                                                                                                                                       |
| 2022 & 2023 |  UMD Graduate Student Travel Fellowship (twice)                                                                                                                                                                                                                                                                           |
| 2022        |  Rutgers University Travel Award for Conference on Advances in Bayesian and Frequentist Statistics                                                                                                                                                                                                                        |
| 2021 - 2024 |  UMD Tuition Award for External Fellowship (4 times)                                                                                                                                                                                                                                                                      |
|             |  NIH Predoctoral Fellowship (4 times) <a href="#">[about NIH Graduate Partnerships Program (GPP)]</a>                                                                                                                                                                                                                     |
| 2019 & 2020 |  UMD Dean’s Fellowship (twice)                                                                                                                                                                                                                                                                                            |
| 2017        |  First Award in Thesis Contest on Deep Integration of Key Competence and Classroom Teaching (NO.1 Middle School Affiliated to Central China Normal University)                                                                                                                                                            |
| 2016        |  USTC Outstanding Graduate                                                                                                                                                                                                                                                                                                |
| 2014 & 2015 |  USTC First Award of Academic Scholarship (twice)                                                                                                                                                                                                                                                                         |
| 2014        |  NNU Outstanding Graduate                                                                                                                                                                                                                                                                                                 |

## Honors (continued)

|             |   |                                                                                                                                                                  |
|-------------|---|------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|             | ■ | NNU First Award of Feng Ruer Scholarship                                                                                                                         |
|             | ■ | NNU Outstanding Dissertation                                                                                                                                     |
| 2013        | ■ | Certificate of Recommendation for Exemption from Entrance Examination—USTC Graduate Program                                                                      |
|             | ■ | Eligibility for Recommended Admission (top 5% in Mathematics Teacher Education Program, NNU)                                                                     |
|             | ■ | Second Award of Jiangsu Teaching Professional Skills Contest (ranked 4th in Mathematics across the Jiangsu Province)                                             |
| 2012        | ■ | NNU Second Award in Mathematical Modeling Contest                                                                                                                |
|             | ■ | NNU Outstanding Debate Team Award, Debate Competition on “Is Beauty Subjective or Objective?” Member of the award-winning team arguing that beauty is objective. |
| 2010 & 2011 | ■ | NNU First Award of Rolling Scholarship (twice)                                                                                                                   |
| 2009        | ■ | NNU First Award in Computer Contest                                                                                                                              |

## Academic Activities

### Invited Presentations

|         |   |                                                                                                                           |
|---------|---|---------------------------------------------------------------------------------------------------------------------------|
| 2026/07 | ■ | International Biometric Conference (IBC), Seoul, South Korea (Invited Talk)                                               |
| 2026/05 | ■ | Biostatistics Seminar, UC Davis Comprehensive Cancer Center, Davis, California, USA (Invited Talk)                        |
|         | ■ | STATGEN 2026: Conference on Statistics in Genomics and Genetics, Emory University, Atlanta, Georgia, USA (Invited Poster) |
|         | ■ | The 10th Annual NIH Vivian Pinn Symposium, Bethesda, Maryland, USA (Invited Poster)                                       |
| 2026/01 | ■ | Biostatistics Branch of NCI-DCEG Works in Progress, Rockville, Maryland, USA (Invited Talk)                               |
| 2024/10 | ■ | Women in Statistics and Data Science (WSDS), Reston, Virginia, USA (Invited Panelist)                                     |

### Contributed Presentations

|      |   |                                                                                                                                                                                                                 |
|------|---|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2026 | ■ | The American Society of Human Genetics (ASHG) Annual Meeting, Montreal, Canada                                                                                                                                  |
|      | ■ | Joint Statistical Meetings (JSM), Boston, Massachusetts, USA                                                                                                                                                    |
|      | ■ | The 11th Workshop on Biostatistics and Bioinformatics, Georgia State University, Atlanta, Georgia, USA (Contributed Poster as a recipient of the IMS New Researchers Travel Award and NSF Travel Support Award) |
|      | ■ | AI Day for Federal Statistics, National Academy of Sciences, Washington DC, USA                                                                                                                                 |
| 2025 | ■ | Joint Statistical Meetings (JSM), Nashville, Tennessee, USA                                                                                                                                                     |
|      | ■ | The 17th Annual NCI-DCEG Fellows' Training Symposium, Rockville, Maryland, USA                                                                                                                                  |
| 2024 | ■ | Joint Statistical Meetings (JSM), Portland, Oregon, USA                                                                                                                                                         |
|      | ■ | AI Day for Federal Statistics, National Academy of Sciences, Washington DC, USA                                                                                                                                 |
|      | ■ | Probability and Statistics Day, University of Maryland, Baltimore County (UMBC), Baltimore, Maryland, USA                                                                                                       |
|      | ■ | NIH Annual Graduate Student Research Symposium, Bethesda, Maryland, USA                                                                                                                                         |
| 2023 | ■ | Women in Statistics and Data Science (WSDS), Seattle, Washington, USA                                                                                                                                           |
|      | ■ | Joint Statistical Meetings (JSM), Toronto, Canada                                                                                                                                                               |

## Academic Activities (continued)

- 2022
  - Probability and Statistics Day, University of Maryland, Baltimore County (UMBC), Baltimore, Maryland, USA
  - Conference on Advances in Bayesian and Frequentist Statistics with a Celebration of the 80th Birthday of Professor William E. Strawderman, Rutgers University, New Brunswick, New Jersey, USA
  - Annual Conference on Quantitative Approaches in Biology, Northwestern University, Chicago, Ohio, USA

## Attendant

- 2026
  - The 31st CBA Annual Conference, Gaithersburg, Maryland, USA
  - Artificial Intelligence (AI) Expo, The Department of Health and Human Services (HHS), Washington D.C., USA
  - NIH Library Artificial Intelligence (AI) Trainings led by OpenAI experts, Bethesda, Maryland, USA
  - Biostatistics Symposium of South California (BSSC), Newport Beach, California, USA
- 2025
  - Eastern North American Region (ENAR), New Orleans, Louisiana, USA
- 2024
  - Statistics in the Age of AI Conference, The George Washington University (GWU), Washington DC, USA
  - Eastern North American Region (ENAR), Baltimore, Maryland, USA
- 2023
  - National Heart, Lung, and Blood Institute (NHLBI) Biostatistics Workshop on Clinical Trial Designs: Innovative Endpoints and Futility Monitoring, Bethesda, Maryland, USA
- 2022
  - International Small Area Estimation (SAE) Annual Conference, UMD, College Park, Maryland, USA
- 2016
  - National Tianyuan Foundation Summer School on Pure Mathematics, Summer Seminar on Pure Mathematics, Zhejiang University, Hangzhou, China
- 2013
  - Yangtze River Delta Symposium on Partial Differential Equations and Doctoral Student Forum, Tongji University, Shanghai, China

## Grant Writing Workshop

- 2026
  - NCI K Grant Writing Workshop, Bethesda, Maryland, USA
  - NCI-DCEG Writing Your Specific Aims Session, Rockville, Maryland, USA
- 2025
  - NCI-DCEG Grant Writing Workshop, Rockville, Maryland, USA

## Software

### R Packages and Research Code

- **Heritability-Velocity** — Author and Maintainer. Reproducible framework for joint estimation of static and dynamic heritability components in longitudinal phenotypes, featuring scalable AI-REML and REHE algorithms for genome-wide data. [\[GitHub\]](#) | [\[arXiv\]](#)
- **PLCO** — Author and Maintainer. Mixed-effects modeling framework with crossed genetic and individual-specific random effects to quantify genetic influences on PSA trajectories in large-scale prostate cancer screening studies. [\[GitHub\]](#) | [\[Paper\]](#)

## Software (continued)

- **AARP** — Author and Maintainer. Implementation of measurement error models for bias correction in food frequency questionnaire-based dietary intake assessment. [\[GitHub\]](#) | [\[Paper\]](#)

## Programming and Statistical Computing

- R | Python | C/C++ | SQL | Jupyter Notebook

## Research Infrastructure

- NIH Biowulf High-Performance Computing Cluster | All of Us Research Hub | UK Biobank

## Scientific Writing and Reproducibility

- $\text{\LaTeX}$  | Microsoft Word | R Markdown | Git

## Teaching

### Teaching Assistant / Grader

- |                         |                                                                                                                   |
|-------------------------|-------------------------------------------------------------------------------------------------------------------|
| Fall 2019               | ■ <b>Sampling Theory (STAT440)</b> , Undergraduate/master-level.<br><i>Grader, UMD</i>                            |
| Spring 2020             | ■ <b>Introduction to Linear Algebra (MATH240)</b> , Undergraduate-level.<br><i>Teaching Assistant, UMD</i>        |
| Summer 2020             | ■ <b>Introduction to Mathematical Proof (MATH310)</b> , Undergraduate-level.<br><i>Grader, UMD</i>                |
|                         | ■ <b>Introduction to Probability Theory (STAT410)</b> , Undergraduate/master-level.<br><i>Grader, UMD</i>         |
| Fall 2020 & Spring 2021 | ■ <b>Elementary Probability and Statistics (STAT100)</b> , Undergraduate-level.<br><i>Teaching Assistant, UMD</i> |
| Spring 2015 & 2016      | ■ <b>Calculus II</b> , Undergraduate-level.<br><i>Teaching Assistant, USTC</i>                                    |
| Fall 2016               | ■ <b>Calculus I</b> , Undergraduate-level.<br><i>Teaching Assistant, USTC</i>                                     |

### Mathematics Teacher (Internship)

- |                   |                                                                                                                                 |
|-------------------|---------------------------------------------------------------------------------------------------------------------------------|
| 2013/09 - 2013/11 | ■ <b>High School Mathematics</b> . High school level.<br><i>Mathematics Teacher Intern, Jinling High School, Nanjing, China</i> |
|-------------------|---------------------------------------------------------------------------------------------------------------------------------|

## Service

### Journal Reviewer

- |                |                                                                                                                                                                                  |
|----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2021 - present | ■ Biometrics (1); Statistics in Medicine (2); BMC Medical Research Methodology (1); Journal of Biopharmaceutical Statistics (1); Frontiers in Oncology (1); LabMed Discovery (1) |
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## Service (continued)

### Judging

- 2026  Chief Judge, NIH Fellows Award for Research Excellence (FARE) 2027 Competition

### Conference / Seminar Organizer

- 2024  Co-organizer of the panel "Discussion of Mentoring Program and Participant Experiences" at WSDS 2024
-  Chair of the session "Rethinking Current Modeling Strategies: Advances and Applications" at JSM 2024
- 2023  Chair of the session "Concurrent - Statistical Insights into Mortality and Inference: From Choquet Capacities to Demographic Patterns" at WSDS 2023

### Mentoring

- 2025–2026  Co-mentor (with **Paul S. Albert, PhD**) of postbaccalaureate fellow **Sumaja Bandreddi** (M.S., B.A., University of Virginia), Biostatistics Branch, DCEG, NCI, NIH
- Spring 2021  Directed Reading Program mentor for undergraduate **Andrew Craun**, Department of Mathematics, University of Maryland, College Park

### Leadership

- 2026  Member, NIH Fellows Award for Research Excellence (FARE) 2027 Committee
- 2022–2023  Vice President, Women in Mathematics, University of Maryland, College Park
- 2020–2021  Representative, Graduate Student Government, University of Maryland, College Park
- 2010–2011  Team Leader, Volunteer Interpreters Group, Nanjing Yunjin Museum, Nanjing, China

### Volunteer Service

- 2026  Volunteer, The Chinese Biopharmaceutical Association-USA (CBA)
- 2025  Volunteer, Sino-American Pharmaceutical Professionals Association (SAPA)-DC
- 2022  Volunteer Interviewee, The Center for Mathematics Education, University of Maryland, College Park
- 2009–2010  Volunteer Teacher, Nanjing Hongshan Primary School for Children of Migrant Workers, Nanjing, China

## Memberships

- 2026 - present  American Society of Human Genetics
- 2024 - present  Washington Statistical Society
-  International Chinese Statistical Association
-  Section of Statistics in Genetics and Genomics
- 2023 - present  American Statistical Association
-  Institute of Mathematical Statistics
-  Society for Mathematical Biology
-  Caucus for Women in Statistics and Data Science
-  Eastern North America Region/International Biometric Society
- 2019 - 2025  American Mathematical Society

## References

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### **Paul S. Albert**

Senior Investigator

Director, Biostatistics Branch

Division of Genetic Epidemiology & Genetics

National Cancer Institute

Rockville, MD 20850, USA

[albertp@mail.nih.gov](mailto:albertp@mail.nih.gov)

### **Jianxin Shi**

Senior Investigator

Biostatistics Branch

Division of Genetic Epidemiology & Genetics

National Cancer Institute

Rockville, MD 20850, USA

[jianxin.shi@nih.gov](mailto:jianxin.shi@nih.gov)

### **Doron Levy**

Professor of Mathematics

Chair, Department of Mathematics

Director, Brin Mathematics Research Center

Co-director, the NCI-UMD Partnership for Integrative Cancer Research

Fellow of the Society for Industrial and Applied Mathematics (SIAM)

University of Maryland, College Park

College Park, MD 20740, USA

[dlevy@umd.edu](mailto:dlevy@umd.edu)